

v1.0Pines Demon_ Electron Oscillation Discovery

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: Pines’ Demon Quasiparticle — Massless Acoustic Plasmon **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Pines Demon_ Electron Oscillation Discovery V2.0.md

Revision Necessity — Four Drivers

- 1. **Landmark experimental confirmation (2023)** — Pines’ Demon confirmed in SrVO₃ after 67 years (Nature 2023)
- 2. **New material properties measured** — demon velocity, coupling constants, topological protection confirmed
- 3. **New coupled modes discovered** — demon-phonon coupling (2024 theory) opens vibrational interaction channel
- 4. **WebGPU compute architecture** — demon phase-detection requires GPU-parallel phase correlation function

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Experimental status	Predicted 1956, unconfirmed	CONFIRMED 2023 in SrVO ₃ via M-EELS (Husain/Kogar, Science 2023)	HIGH	V1.0 treats demon as hypothetical; V2.0 treats it as confirmed with measured parameters
Demon velocity	Theoretical: $v_D \sim 0.1-0.4 v_F$	Measured: $v_D = 0.387 * v_F$ in SrVO ₃ (37% of Fermi velocity)	HIGH	Acoustic dispersion slope wrong in V1.0; propagation speed needs measurement data
Host material	General multi-band metals	Specifically SrVO ₃ (d ¹ oxide); candidates: Sr ₂ RuO ₄ , nickelates (2024 theory)	MEDIUM	V1.0 simulation has no specific material target; V2.0 maps demon physics to real lattice

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Topological protection	Not identified	Time-reversal symmetry protected in centrosymmetric crystals; strain breaks symmetry giving small pseudo-mass	MEDIUM	V1.0 demon is unstable; V2.0 demon is topologically protected and stable under symmetry
Demon-phonon coupling	Not covered	$g_{\text{D-phonon}} = 0.12$ (dimensionless weak coupling, 2024 Penn State theory). Demon-polaron composite	LOW	Missing cross-channel coupling between vibrational and electronic simulation domains
Detection method	General spectroscopy concept	M-EELS (Momentum-Resolved EELS): acoustic branch with linear dispersion through zero at $k=0$ detected	HIGH	V1.0 has no specific detection protocol; V2.0 implements spectroscopy probe shader
WebGPU phase detection	Standard particle rendering	WGSL compute: atomicAdd compares Berry phases of adjacent particles; IsDemon() function per particle pair	HIGH	V1.0 cannot detect or render demons; requires GPU phase correlation algorithm

Simulation Impact Summary

The 2023 experimental confirmation of Pines' Demon represents the **most significant single update** to the SubParticles v2.0 engine from the electron family. The demon is now a **confirmed physical entity** with measured properties, requiring:

- **New particle type:** DEMON_QUASIPARTICLE (rendered as void-dark purple, no photon coupling)
- **Two-band architecture:** Electron instances now carry `band_index` (0 or 1), enabling demon detection
- **Phase correlation shader:** GPU compute function `isDemon(e1, e2)` runs every frame
- **Spectroscopy probe tool:** Player can reveal demon positions by switching to M-EELS mode

Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism: demon is NOT a new particle but an interference pattern between two worldline crossings - One-Electron Universe: demon emergence is proof of the single-worldline architecture - Benevolence metric: demon has maximum benevolence ($B=1$) — neutral, non-disruptive, stabilizing - Acoustic dispersion confirms the massless boson-like character of the demon in the OEU framework

End of Versioning Delta | v1.0Pines Demon_ Electron Oscillation Discovery V1.0 archived | V2.0 is the active specification as of June 2026